

# Companion to Bird & de Moor, Algebra of Programming

## §10. Greedy Algorithms

### §10.1. Theory

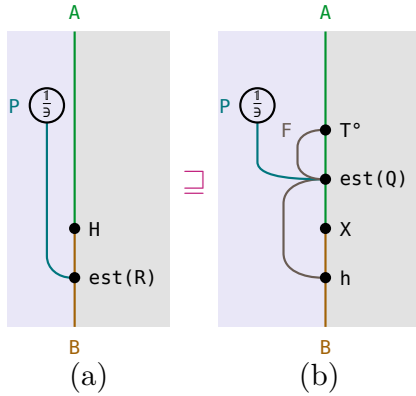
(10.1a)

$h, T, R, H, M$  as in  $\triangle H$ ; additionally  $Q$  a **connected** preorder on the sets  $\frac{T^\circ}{\exists}$  returns, so that  $\frac{T^\circ}{\exists} \text{ est}(Q)$  is entire.

**Map(h)  $\wedge$  MonoAlg(h,R)  $\wedge$  R R  $\subseteq$  R  $\wedge$  ThinCondition(T,h,R,Q)  $\Rightarrow$  ( $\mu X : \frac{T^\circ}{\exists} \text{ est}(Q) F(X)$  Theorem 10.1  
 $h) \subseteq \frac{H}{\exists} \text{ est}(R)$**

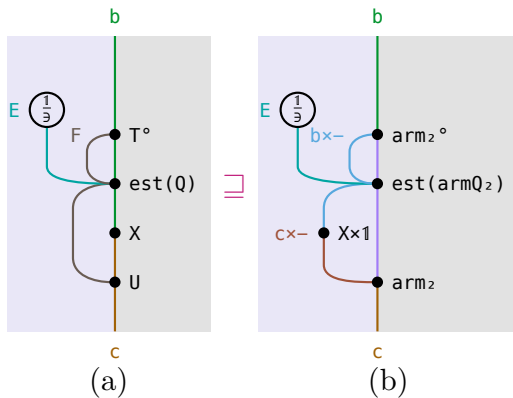
the same optimum reached by keeping ONE decomposition at each step, so that no set is ever carried  
 and the recursion runs on values alone

$H \triangle (T)^\circ (h) \triangle H, Q$



|     |                                                             |  |
|-----|-------------------------------------------------------------|--|
| (a) | $\frac{H}{\exists} \text{ est}(R)$                          |  |
| (b) | $\sqsubseteq \frac{T^\circ}{\exists} \text{ est}(Q) F(X) h$ |  |

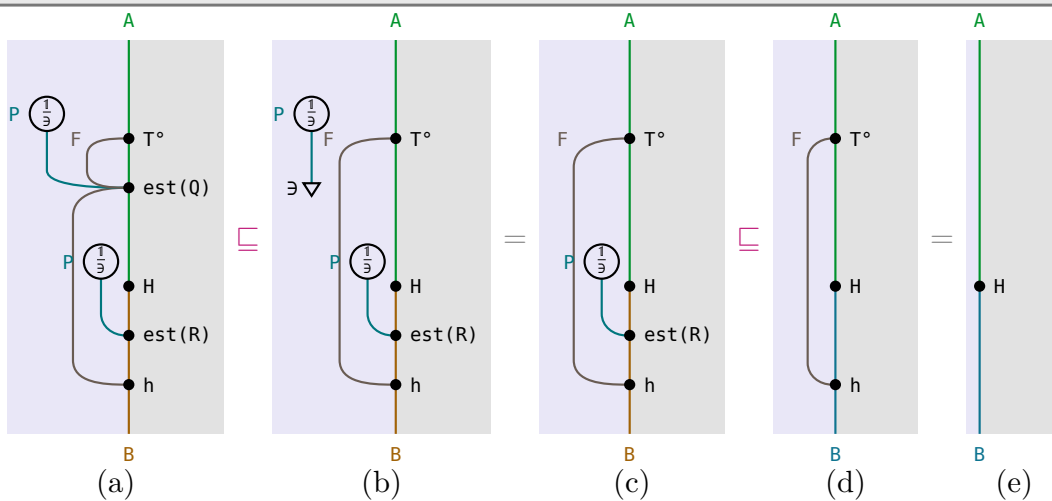
the body at  $T$  a coproduct of two arms



|     |                                                                                                       |                                               |
|-----|-------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| (a) | $\frac{T^\circ}{\exists} \text{ est}(Q) F(X) U$                                                       | Proposition 10.1, $V_2 V_1^\circ = \emptyset$ |
| (b) | $\sqsubseteq \frac{\text{arm}_2^\circ}{\exists} \text{ est}(\text{arm}_2) (X \times 1) \text{ arm}_2$ |                                               |

**$T^\circ F(H) h = H \Rightarrow \frac{T^\circ}{\exists} \text{ est}(Q) F(\frac{H}{\exists} \text{ est}(R)) h \subseteq H$**  (10.1b)

taking the input apart in one  $Q$ -extreme way, solving the parts by  $M$  and assembling by  $h$  returns only what  $H$   
 returns

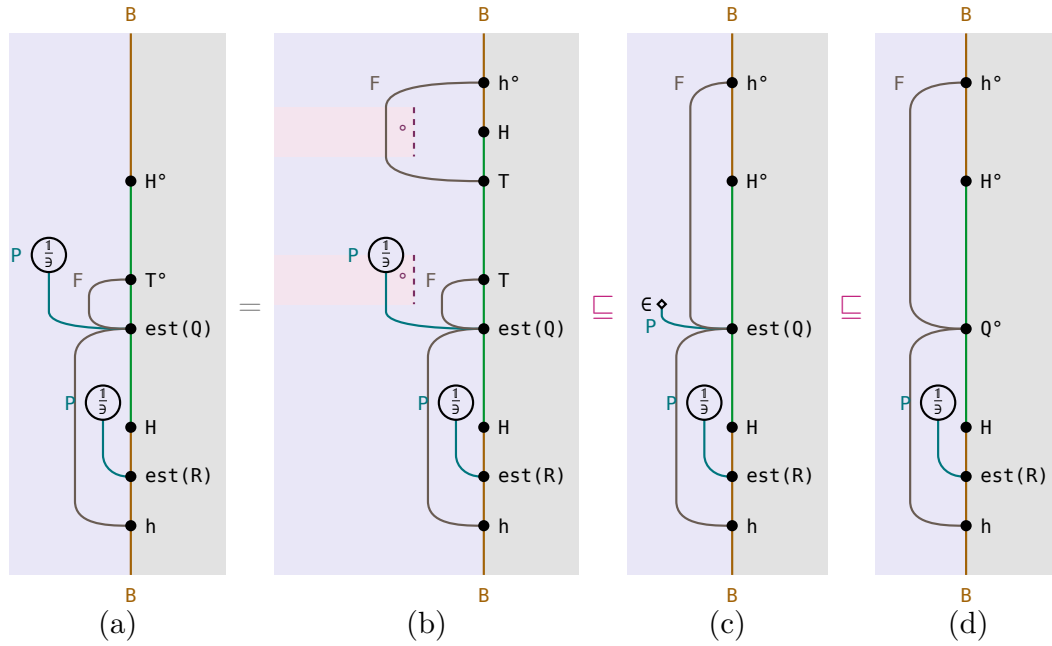


|     |                                                                                     |                                                |
|-----|-------------------------------------------------------------------------------------|------------------------------------------------|
| (a) | $\frac{T^\circ}{\exists} \text{ est}(Q) F(\frac{H}{\exists} \text{ est}(R)) h$      | $\text{est}(Q) \subseteq \exists$              |
| (b) | $\sqsubseteq \frac{T^\circ}{\exists} \exists F(\frac{H}{\exists} \text{ est}(R)) h$ | $\frac{R}{\exists} \exists = R$                |
| (c) | $= T^\circ F(\frac{H}{\exists} \text{ est}(R)) h$                                   | $\frac{R}{\exists} \text{ est}(R) \subseteq S$ |
| (d) | $\sqsubseteq T^\circ F(H) h$                                                        |                                                |
| (e) | $= H$                                                                               |                                                |

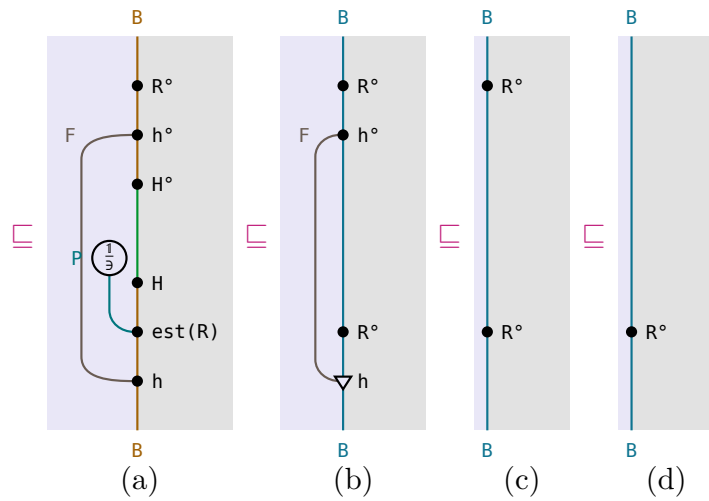
$$\text{Map}(h) \wedge \text{MonoAlg}(h, R^\circ) \wedge R^\circ R^\circ \subseteq R^\circ \wedge T^\circ F(H) h = H \wedge Q F(H) h \subseteq F(H) h R \implies H^\circ \stackrel{T^\circ}{\subseteq} \text{est}(Q) \quad (10.1c)$$

$$F(\stackrel{H}{\subseteq} \text{est}(R)) h \subseteq R^\circ$$

$H^\circ$  followed by the greedy body at  $M$  is  $\subseteq R^\circ$ : an answer of the body is never worse than an answer of  $H$  to the same input



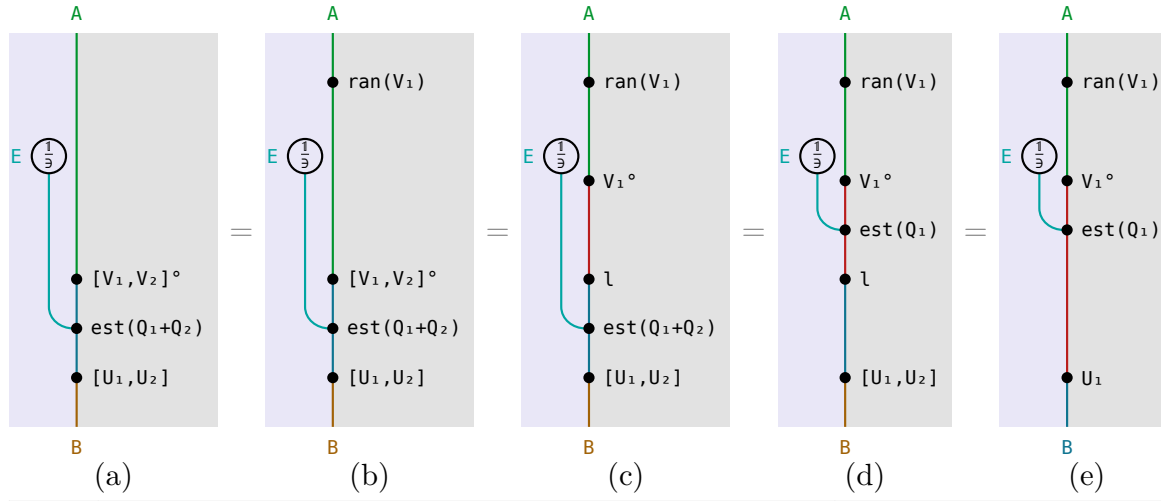
|     |                                                                                                                 |                                                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $H^\circ \stackrel{T^\circ}{\subseteq} \text{est}(Q) F(\stackrel{H}{\subseteq} \text{est}(R)) h$                | $T^\circ F(H) h = H \implies H^\circ = h^\circ F(H^\circ) T$                                                                        |
| (b) | $= h^\circ F(H^\circ) T \stackrel{T^\circ}{\subseteq} \text{est}(Q) F(\stackrel{H}{\subseteq} \text{est}(R)) h$ | $T \stackrel{T^\circ}{\subseteq} \in$                                                                                               |
| (c) | $\subseteq h^\circ F(H^\circ) \in \text{est}(Q) F(\stackrel{H}{\subseteq} \text{est}(R)) h$                     | $\in \text{est}(R) \subseteq R^\circ$                                                                                               |
| (d) | $\subseteq h^\circ F(H^\circ) Q^\circ F(\stackrel{H}{\subseteq} \text{est}(R)) h$                               | $Q F(H) h \subseteq F(H) h R \implies h^\circ F(H^\circ) Q^\circ \subseteq R^\circ h^\circ$<br>$F(H^\circ)$<br>$t(f g) = t(f) t(g)$ |



|     |                                                                                |                                                                   |
|-----|--------------------------------------------------------------------------------|-------------------------------------------------------------------|
| (a) | $\subseteq R^\circ h^\circ F(H^\circ \stackrel{H}{\subseteq} \text{est}(R)) h$ | $S^\circ \stackrel{S}{\subseteq} \text{est}(R) \subseteq R^\circ$ |
| (b) | $\subseteq R^\circ h^\circ F(R^\circ) h$                                       |                                                                   |
| (c) | $\subseteq R^\circ R^\circ$                                                    |                                                                   |
| (d) | $\subseteq R^\circ$                                                            |                                                                   |

$$V_2 \ V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2] = (\text{ran}(V_1) \frac{V_1^\circ}{3} \text{est}(Q_1) U_1) \cup (\text{ran}(V_2) \frac{V_2^\circ}{3} \text{est}(Q_2) U_2) \quad \text{Proposition 10.1}$$

when  $V_1$  and  $V_2$  have disjoint ranges, the greedy step over  $[V_1, V_2]$  runs the  $V_1$  step on inputs  $V_1$  reaches and the  $V_2$  step on inputs  $V_2$  reaches



|     |                                                                                                                                                                  |                                                                                                     |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| (a) | $\frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2]$                                                                                                    | empty off $\text{ran } V_1 \cup \text{ran } V_2$                                                    |
| (b) | $= (\text{ran}(V_1) \frac{V_1^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \frac{V_2^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2])$           | $\frac{[V_1, V_2]^\circ}{3} = \frac{V_1^\circ}{3} P(\text{inl})$ on $\text{ran } V_1$ — dp-disjoint |
| (c) | $= (\text{ran}(V_1) \frac{V_1^\circ}{3} P(l) \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \frac{V_2^\circ}{3} P(r) \text{est}(Q_1 + Q_2) [U_1, U_2])$ | $P(\text{inl}) \text{est}(Q_1 + Q_2) = \text{est}(Q_1) \text{inl}$                                  |
| (d) | $= (\text{ran}(V_1) \frac{V_1^\circ}{3} \text{est}(Q_1) l [U_1, U_2]) \cup (\text{ran}(V_2) \frac{V_2^\circ}{3} \text{est}(Q_2) r [U_1, U_2])$                   | $\text{inl}[U_1, U_2] = U_1$                                                                        |
| (e) | $= (\text{ran}(V_1) \frac{V_1^\circ}{3} \text{est}(Q_1) U_1) \cup (\text{ran}(V_2) \frac{V_2^\circ}{3} \text{est}(Q_2) U_2)$                                     |                                                                                                     |

## §10.2. The detab-entab problem

(10.2a)

$\text{detab} \triangleq ([\text{nil}, \text{expand}]) : \text{String} \rightarrow \text{String}$  over  $\text{snoc}$ -lists,  $\alpha \triangleq [\text{nil}, \text{snoc}]$ ,  $H = \text{detab}^\circ$ ;  $\text{expand } (xs, a) = (a = \text{TB} \rightarrow \text{fill } xs, xs \# [a])$ ,  $\text{fill } xs = xs \# \text{blanks } (n - (\text{col } xs) \bmod n)$ .

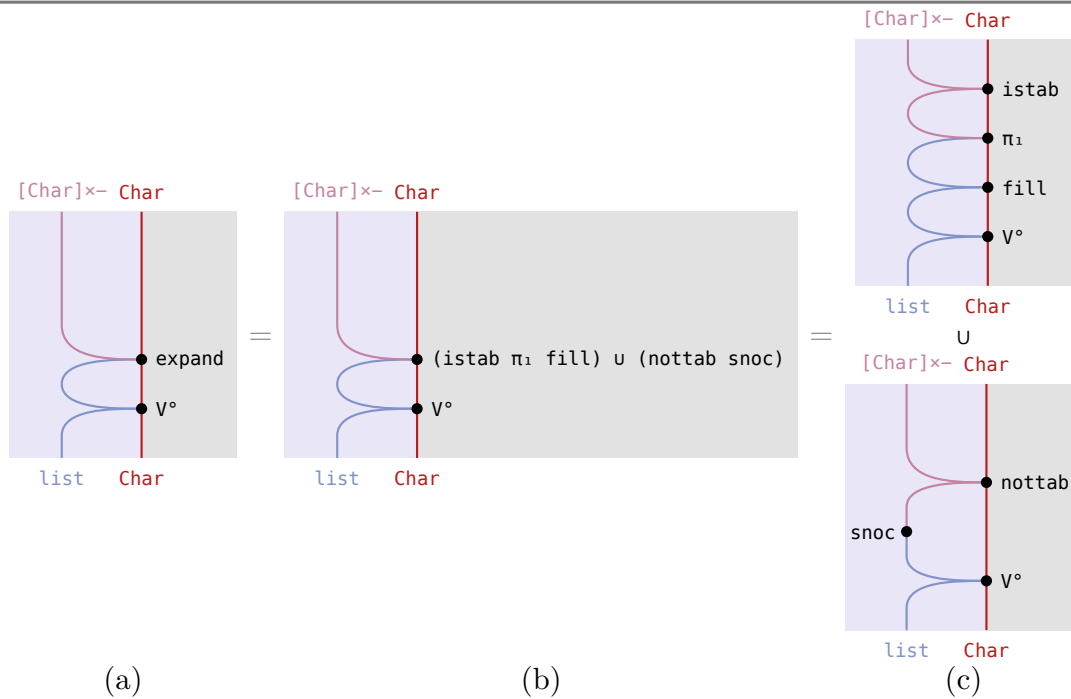
$\text{col} \triangleq ([\text{zero}, \text{count}])$ ,  $\text{count } (c, a) = (a = \text{NL} \rightarrow 0, c + 1)$ ; TB the tab, BL the blank, NL the newline, tab stops every  $n$  columns.

$R \triangleq \text{length} \leq \text{length}^\circ$ ,  $U$  the preorder with  $a \cup b \iff a = \text{TB} v a = b$ ,  $V \triangleq \text{prefix}^\circ n (\text{fill } \text{fill}^\circ)$ ,  $Q \triangleq 1 + (V \times U)$ .

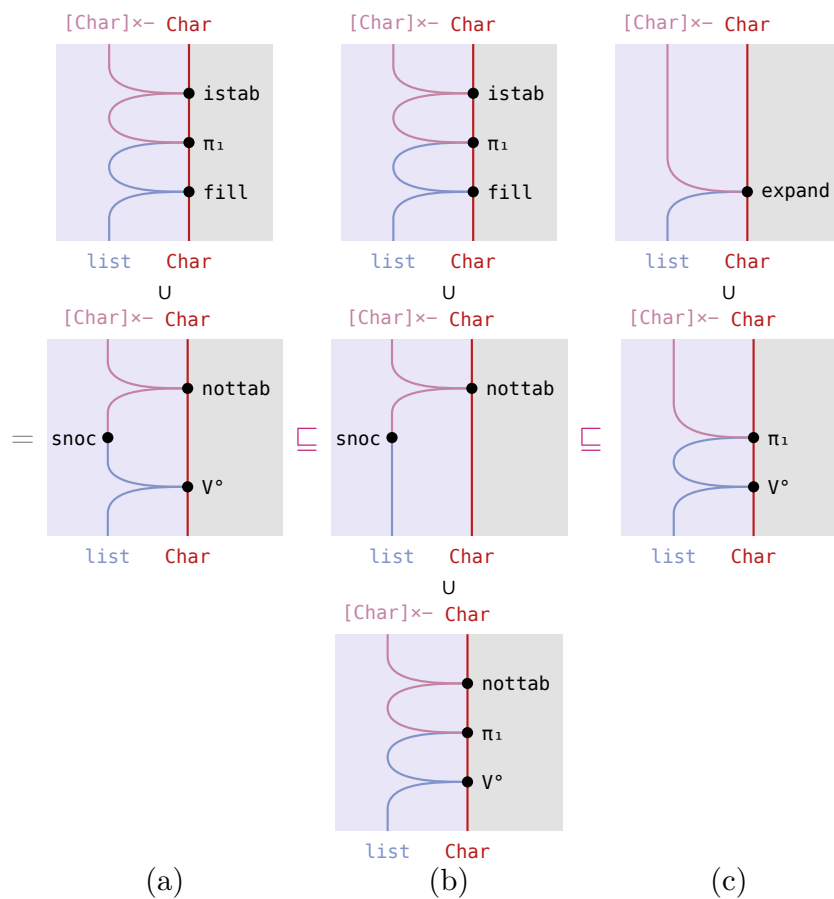
$\text{unfill } xs$  the shortest prefix of  $xs$  with  $\text{fill}$  ( $\text{unfill } xs = \text{fill } xs$ );  $\text{tbc}$  the trailing blank count,  $\text{triple} \triangleq (\text{unfill } \text{entab}, (\text{tbc}, \text{col}))$ .

$$0 < n \wedge \text{blank} \neq n\text{l} \implies \text{expand } V^\circ \sqsubseteq \text{expand } u (\pi_1 V^\circ) \quad (10.2b)$$

shortening the output of one **expand** step to a  $V$ -smaller string either keeps the whole step (**expand**) or drops its character and shortens its input string ( $\pi_1 V^\circ$ )



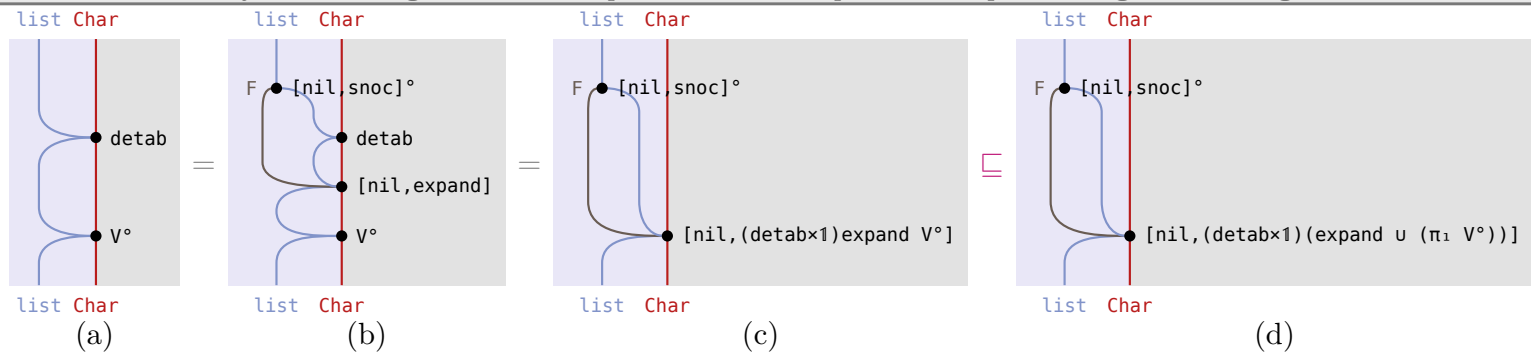
|     |                                                                                    |                                                                              |
|-----|------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| (a) | $\text{expand } V^\circ$                                                           | $\text{expand} = \text{istab outl fillunottab snocR}$ — definition of expand |
| (b) | $= ((\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab snoc})) V^\circ$         | $(\text{SuT})R = \text{SRuTR}$ — conditionals distribute                     |
| (c) | $= (\text{istab } \pi_1 \text{ fill } V^\circ) \cup (\text{nottab snoc } V^\circ)$ | $\text{fill } V^\circ = \text{fill}$ (Ex. 10.4)                              |



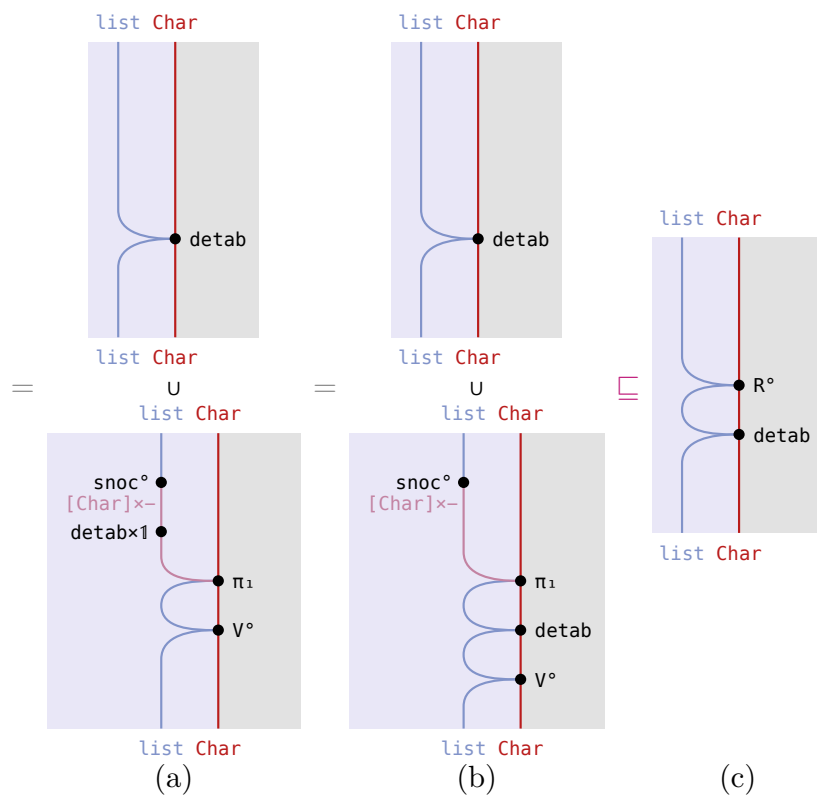
|     |                                                                                                    |                                                                                                                                |
|-----|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| (a) | $= (\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab snoc } V^\circ)$                          | $\text{snoc } V^\circ \sqsubseteq \text{snoc} \cup (\pi_1 V^\circ)$ (Ex. 10.4)                                                 |
| (b) | $\sqsubseteq (\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab (snoc } \cup (\pi_1 V^\circ)))$ | $\text{expand} = \text{istab outl fillunottab snocR}$ — definition of expand; $\text{nottab } R \sqsubseteq R$ — guard dropped |
| (c) | $\sqsubseteq \text{expand } u (\pi_1 V^\circ)$                                                     |                                                                                                                                |

$$0 < n \wedge \text{blank} \neq n\mathbf{l} \Rightarrow \text{detab } V^\circ \sqsubseteq R^\circ \text{ detab}$$

(10.2c)

any  $V$ -shortening of **detab**'s output is **detab**'s output on an input no longer than the given one

|     |                                                                                                              |                                                                                                                             |
|-----|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{detab } V^\circ$                                                                                      | $\text{detab}$ is a fold: $\text{detab} = \alpha^\circ F(\text{detab}) [\text{nil}, \text{expand}] \triangleq \text{detab}$ |
| (b) | $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand}] V^\circ$                      | coproducts, and $\text{nil } V^\circ = \text{nil}$ (Exercise 10.4)                                                          |
| (c) | $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } V^\circ]$                     | the claim — (10.2b)                                                                                                         |
| (d) | $\sqsubseteq [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } u (\pi_1 V^\circ)]$ | distributing $u$ ; the fold again, and the definition of $F$                                                                |

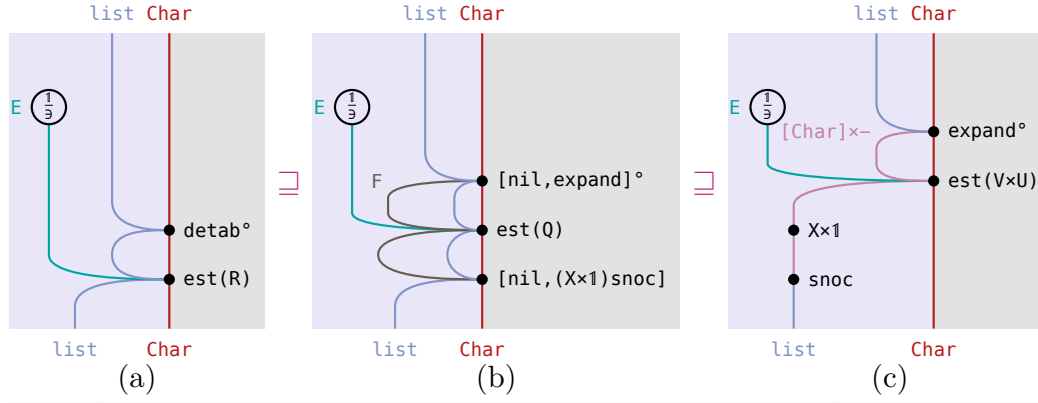


|     |                                                                               |                                                                                                                                                                                                                                                         |
|-----|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $= \text{detab } u (\text{snoc}^\circ (\text{detab} \times 1) \pi_1 V^\circ)$ | naturality of $\pi_1$ : $(\text{detab} \times 1) \pi_1 = \pi_1 \text{ detab}$ ; $\text{snoc}^\circ \pi_1$ is $\text{init}$                                                                                                                              |
| (b) | $= \text{detab } u (\text{snoc}^\circ \pi_1 \text{ detab } V^\circ)$          | $\text{init}$ is inductive, so $X \triangleq \text{detab } V^\circ$ , a solution of $X \sqsubseteq \text{detab } u (\text{init } X)$ , lies below the greatest one, $\text{prefix detab}$ (induction on the input); $\text{prefix} \sqsubseteq R^\circ$ |
| (c) | $\sqsubseteq R^\circ \text{ detab}$                                           |                                                                                                                                                                                                                                                         |

$$0 < n \wedge \text{blank} \neq n1 \Rightarrow (\mu X : \frac{[\text{nil}, \text{expand}]^\circ}{3} \text{est}(Q) F(X) [\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{detab}^\circ}{3} \text{est}(R) \quad (10.2d)$$

the shortest input **detab** expands to the given output is one pass along that output, holding each blank back and cashing the held blanks in for a tab wherever the column reaches a tab stop

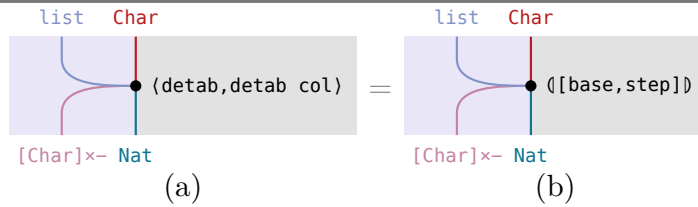
the specification —  $\triangleq \text{detab}; \text{detab entab} = 1$  and nothing shorter does



|     |                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{\text{detab}^\circ}{3} \text{est}(R)$                                                           | at $Q \triangleq 1 + (V \times U)$ : one character of input is decided at each step. $F(\tau, R) \alpha \alpha R$ . $\text{detab prefix} \sqsubseteq R^\circ$ $\text{detab}$ is FALSE, — at $n=8$ , $\text{detab}$ $[a, b, c, d, e, \text{TB}] = [a, b, c, d, e, \text{BL}, \text{BL}, \text{BL}]$ , whose prefix $[a, b, c, d, e, \text{BL}, \text{BL}]$ is longer than any input giving it, and $\text{detab } V^\circ \sqsubseteq R^\circ$ $\text{detab}$ holds. $\text{expand } V^\circ \sqsubseteq \text{expand } U$ ( $\pi_1 V^\circ$ ) — shortening the output either leaves the last step alone or discards it |
| (b) | $\sqsubseteq \frac{[\text{nil}, \text{expand}]^\circ}{3} \text{est}(Q) F(X) [\text{nil}, \text{snoc}]$ | Proposition 10.1: <b>nil</b> and <b>expand</b> have disjoint ranges. The greedy step is to emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| (c) | $\sqsubseteq \frac{\text{expand}^\circ}{3} \text{est}(V \times U) (X \times 1) \text{snoc}$            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

$$\text{blank} \neq n1 \wedge \text{tb} \neq n1 \Rightarrow \langle \text{detab}, \text{detab col} \rangle = \langle [\text{base}, \text{step}] \rangle \quad (10.2e)$$

one pass over the input carries the output so far together with its column



|     |                                                  |                                                                                                                                                                                                  |
|-----|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\langle \text{detab}, \text{detab col} \rangle$ | both components are folds over the same input; <b>step</b> on $(\text{detab } x, \text{col}(\text{detab } x))$ is $(\text{detab}, \text{col detab})$ one character further — not a tabulated row |
| (b) | $= \langle [\text{base}, \text{step}] \rangle$   |                                                                                                                                                                                                  |

Exercise 10.1

$$\text{blank} \neq n1 \wedge \text{tb} \neq n1 \Rightarrow \langle [\text{base}, \text{step}] \rangle (\text{convert}(l), r) \Leftrightarrow r = \text{loop}(\text{step}, (([]), 0), l)$$

the fold of  $[\text{base}, \text{step}]$  over the input converted to a snoc-list is the left loop of **step** from **base**

$$\pi_1(\text{loop}(\text{step}, (([]), c_0), x)) = \text{loop}'((f, g), (c_0, x))$$

Exercise 10.1: when **step** only appends  $f(c, a)$  to the output and moves the state by  $g$ , the output of the loop is  $\text{loop}'(f, g)$ , which never carries the output

$$a \neq \text{tb} \Rightarrow \text{expand}(y, b) = \text{snoc}(x, a) \Leftrightarrow \text{fill}(y) = \text{snoc}(x, a) \wedge b = \text{tb} \vee (y, b) = (x, a) \quad (10.2f)$$

**expand** produces  $x \# [a]$  from a tab after a string that fills to it, or from  $x$  and  $a$

$$0 < n \wedge \text{blank} \neq n1 \Rightarrow (\exists y. \text{fill}(y) = \text{snoc}(x, a)) \Leftrightarrow a = \text{blank} \wedge \text{col}(\text{snoc } x \text{ } a) \% n = 0$$

a string is a **fill** exactly when it ends in a blank on a tab stop

$$\text{entab}(x) = \text{entab}(\text{unfill } x) ++ \text{blanks}(\text{tbc } x) \quad (10.1)$$

(10.1): the output of **entab** is that of **unfill**  $x$  followed by the trailing blanks held back

$$\text{tbc}[] = 0$$

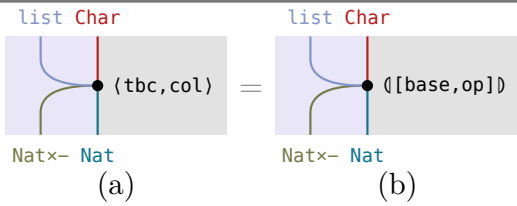
the empty string has no trailing blanks

$$\text{tbc}(\text{snoc}(x, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc } x \text{ } a) \% n \neq 0 \text{ then } \text{tbc } x + 1 \text{ else } 0$$

a blank off a tab stop adds one held blank; anything else releases them all

$$\text{blank} \neq \text{nl} \Rightarrow \langle \text{tbc}, \text{col} \rangle = \langle \llbracket \text{base}, \text{op} \rrbracket \rangle \quad (10.2g)$$

one pass counts the held blanks and the column together



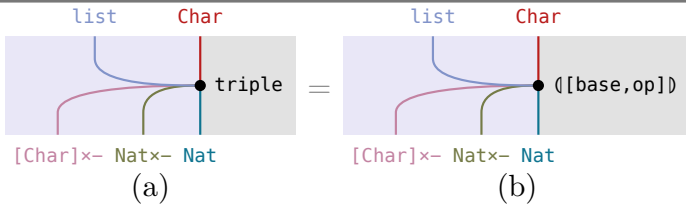
(a)  $\langle \text{tbc}, \text{col} \rangle$

(b)  $= \langle \llbracket \text{base}, \text{op} \rrbracket \rangle$

op on  $\langle \text{tbc } x, \text{col } x \rangle$  is  $\langle \text{tbc}, \text{col} \rangle$  one character further, from the equations for **tbc** — (10.1)

$$\text{blank} \neq \text{nl} \Rightarrow \text{triple} = \langle \llbracket \text{base}, \text{op} \rrbracket \rangle \quad (10.2h)$$

one pass carries the output up to the held blanks, their count and the column



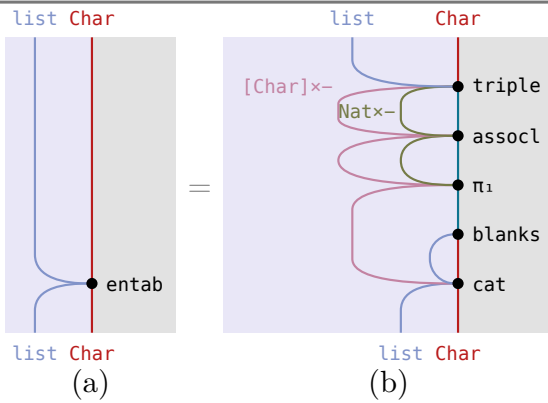
(a) **triple**

(b)  $= \langle \llbracket \text{base}, \text{op} \rrbracket \rangle$

op on **triple**  $x$  is **triple** one character further: (10.1) where the blanks are released — (10.1)

$$\text{entab} = \text{triple assocl } \pi_1 (1 \times \text{blanks}) \text{ cat} \quad (10.2i)$$

**entab** is **triple** with the held blanks appended to its output



(a) **entab**

(b)  $= \text{triple assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$

(10.1) — (10.1)

### §10.3. The minimum tardiness problem

(10.3a)

$\text{FX} = 1 + (X \times \text{Job})$ ,  $\alpha \triangleq [\text{nil}, \text{snoc}]$  on schedules,  $\beta \triangleq [\text{nil}, \text{snag}]$  on bags, **snag** putting a job into a bag; **bagify**  $\triangleq \langle \beta \rangle : [\text{Job}] \rightarrow \text{Bag Job}$ ,  $\text{H} = \text{bagify}^\circ$ .

**ct**, **dt**, **wt** :  $\text{Job} \rightarrow \text{Real}$  the completion, due and weighting quantities of a job; **penalty**(**xs**, **j**) = ( $\text{sum}(\text{list}(\text{ct})(\text{xs})) + \text{ct}(\text{j}) - \text{dt}(\text{j})$ )  $\times$  **wt**(**j**).

$\text{cost} \triangleq \frac{\text{prefix}}{\sup} P(\alpha^\circ [\text{zero}, \text{penalty}]) \text{ est}(\geq)$ ,  $\text{cost} [] = 0$ ,  $\text{cost} (\text{xs} \# [\text{j}]) = \text{bmax} (\text{cost } \text{xs}, \text{penalty } (\text{xs}, \text{j}))$ ,  $\text{R} \triangleq \text{cost} \leq \text{cost}^\circ$ .

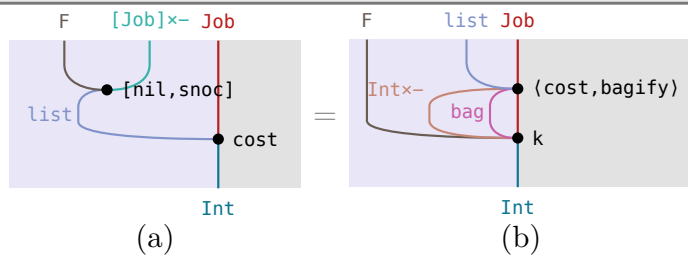
$\text{perm} \triangleq \text{bagify } \text{bagify}^\circ = \langle [\text{nil}, \text{add}] \rangle$ ,  $\text{add} (\text{xs}, \text{j}) = \text{ys} \# [\text{j}] \# \text{zs}$  for some  $\text{xs} = \text{ys} \# \text{zs}$ .

$\text{k} \triangleq [\text{zero}, \text{assocr } (1 \times ((\text{bagify}^\circ \times 1) \text{ penalty})) \text{ bmax}]$ ,  $\text{f} \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{ penalty}]$ ,  $\text{Q} \triangleq \text{f} \leq \text{f}^\circ$ ,  $\text{Q}' \triangleq (\text{bagify}^\circ \times 1) \text{ penalty} \leq \text{penalty}^\circ (\text{bagify} \times 1)$ .

$\text{g} \triangleq [\text{zero}, \text{penalty}]$  (10.5),  $\text{m} \triangleq [\text{zero}, \pi_1 \text{ cost}]$  (B&dM's **h**, renamed as in (10.3e)) (10.6).

$$[\text{nil}, \text{snoc}] \text{ cost} = F(\langle \text{cost}, \text{bagify} \rangle) k \quad (10.3b)$$

the cost of a schedule is got from the cost and the bag of the schedule before its last job, and that job

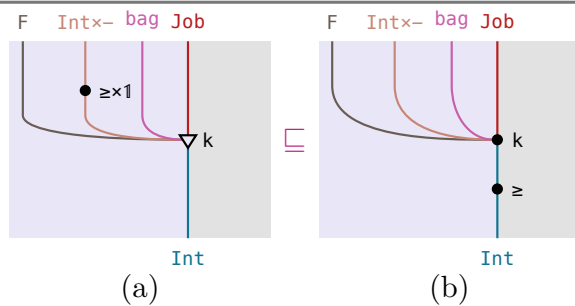


|     |                                                     |                                                                                                                                                                         |
|-----|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $[\text{nil}, \text{snoc}] \text{ cost}$            | $\text{cost}(xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$ ,<br>and $\text{penalty}(xs, j)$ reads only the bag of $xs$ — $\triangleq \text{bagify}$ |
| (b) | $= F(\langle \text{cost}, \text{bagify} \rangle) k$ |                                                                                                                                                                         |

$$F(\geq \times 1) k \sqsubseteq k \geq$$

Exercise 10.5

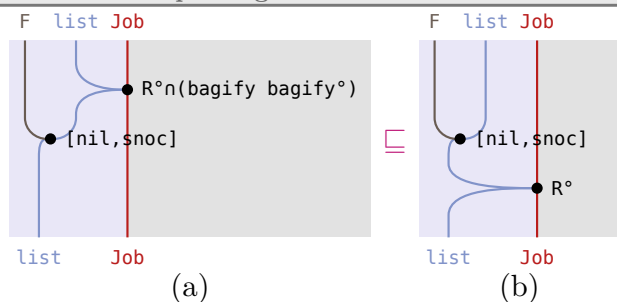
a larger cost before the last job gives a larger cost after it



|     |                      |                                                 |
|-----|----------------------|-------------------------------------------------|
| (a) | $F(\geq \times 1) k$ | $\text{bmax}$ is monotone in its first argument |
| (b) | $\sqsubseteq k \geq$ |                                                 |

$$F(R^\circ \circ n(\text{bagify } \text{bagify}^\circ)) [\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}] R^\circ \quad (10.2)$$

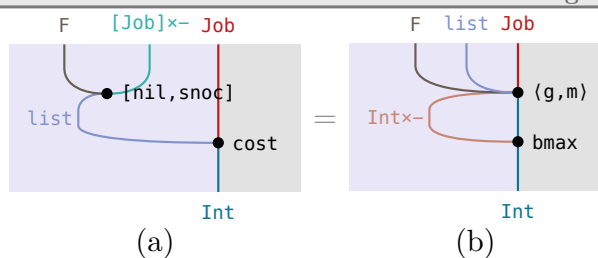
improving the schedule before the last job, among schedules of the same bag, improves the whole schedule



|     |                                                                                    |                                                                                                                       |
|-----|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| (a) | $F(R^\circ \circ n(\text{bagify } \text{bagify}^\circ)) [\text{nil}, \text{snoc}]$ | Proposition 9.3 at $S \triangleq \text{bagify}$ , $\leq \triangleq$ — dp-context-mono, with (10.3b) and Exercise 10.5 |
| (b) | $\sqsubseteq [\text{nil}, \text{snoc}] R^\circ$                                    |                                                                                                                       |

$$[\text{nil}, \text{snoc}] \text{ cost} = \langle g, m \rangle \text{ bmax} \quad (10.4)$$

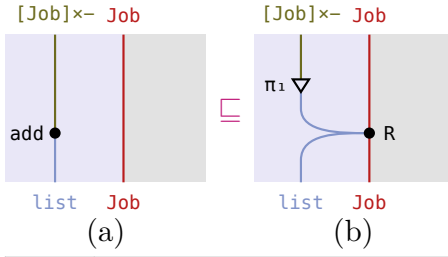
the cost of a schedule is the larger of the penalty of its last job and the cost of the schedule before it



|     |                                          |                                                                                                                                           |
|-----|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $[\text{nil}, \text{snoc}] \text{ cost}$ | $\text{cost}(xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$<br>and $\text{bmax}$ commutes — $\triangleq \text{bagify}$ |
| (b) | $= \langle g, m \rangle \text{ bmax}$    |                                                                                                                                           |

$$(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow add \sqsubseteq \pi_1 R \quad (10.7)$$

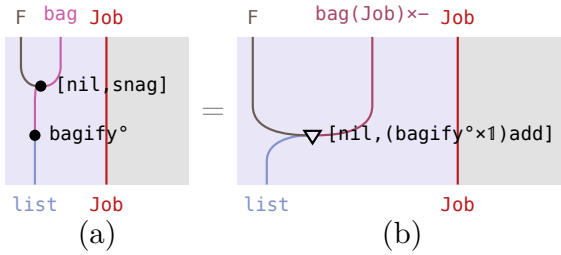
putting one more job anywhere into a schedule never lowers its cost



|     |                       |                                                                                   |
|-----|-----------------------|-----------------------------------------------------------------------------------|
| (a) | add                   | induction on where $j$ goes: every job after it starts later, so no penalty falls |
| (b) | $\sqsubseteq \pi_1 R$ |                                                                                   |

$$[nil, snag] \text{ bagify}^\circ = [nil, (\text{bagify}^\circ \times 1) \text{ add}] \quad (10.8)$$

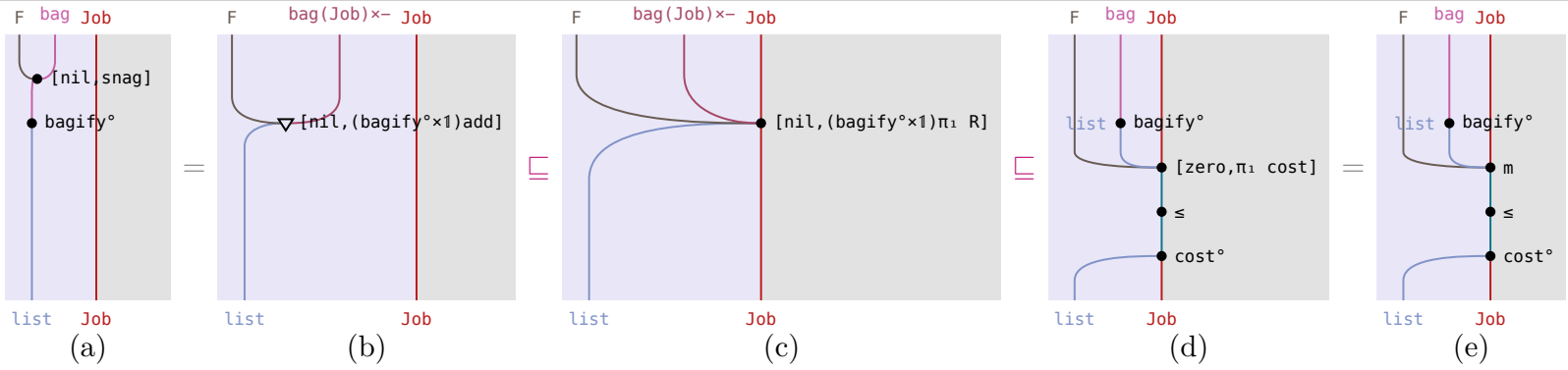
the orderings of a bag with one more job are the orderings of the bag with that job put in anywhere



|     |                                                       |                                                                          |
|-----|-------------------------------------------------------|--------------------------------------------------------------------------|
| (a) | $[nil, snag] \text{ bagify}^\circ$                    | take out the last copy of $j$ and put it back $\triangleq \text{bagify}$ |
| (b) | $= [nil, (\text{bagify}^\circ \times 1) \text{ add}]$ |                                                                          |

$$(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow [nil, snag] \text{ bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) m \leq \text{cost}^\circ \quad (10.3c)$$

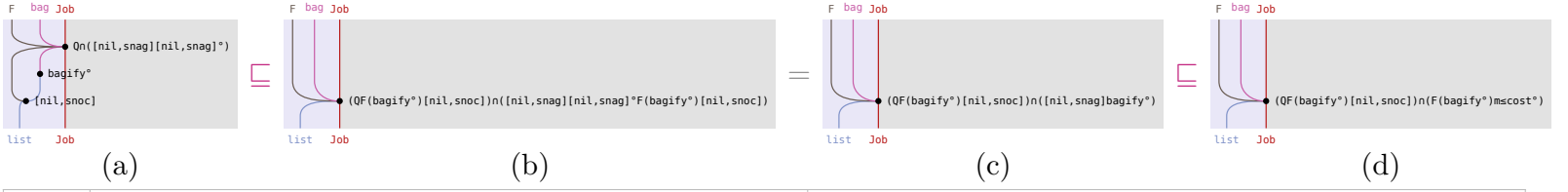
an ordering of a bag with one more job costs at least the cost of the ordering of the bag before its last job



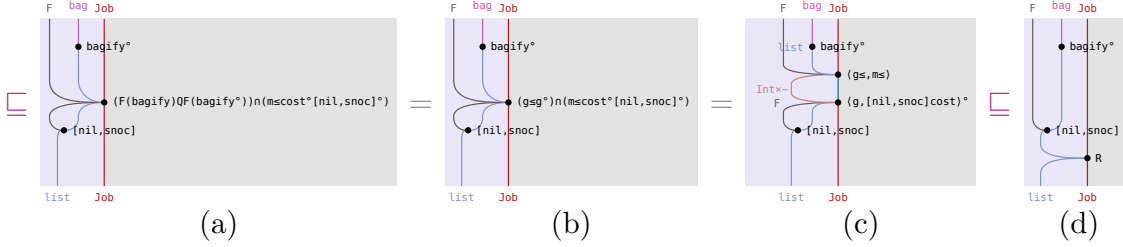
|     |                                                                                        |                                                                              |
|-----|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| (a) | $[nil, snag] \text{ bagify}^\circ$                                                     | (10.8) — (10.8)                                                              |
| (b) | $= [nil, (\text{bagify}^\circ \times 1) \text{ add}]$                                  | (10.7) — (10.7)                                                              |
| (c) | $\sqsubseteq [nil, (\text{bagify}^\circ \times 1) \pi_1 R]$                            | definition of $R$ , and $nil \sqsubseteq \text{zero} \leq \text{cost}^\circ$ |
| (d) | $\sqsubseteq F(\text{bagify}^\circ) [zero, \pi_1 \text{ cost}] \leq \text{cost}^\circ$ | definition of $m$                                                            |
| (e) | $= F(\text{bagify}^\circ) m \leq \text{cost}^\circ$                                    |                                                                              |

$$(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow (Q \ n \ ([nil, snag] \ [nil, snag]^\circ)) \ F(bagify^\circ) \ [nil, snoc] \sqsubseteq F(bagify^\circ) \ [nil, snoc] \ R \quad (10.3)$$

a schedule of a bag ending in a job of least penalty is no worse than any schedule of the same bag



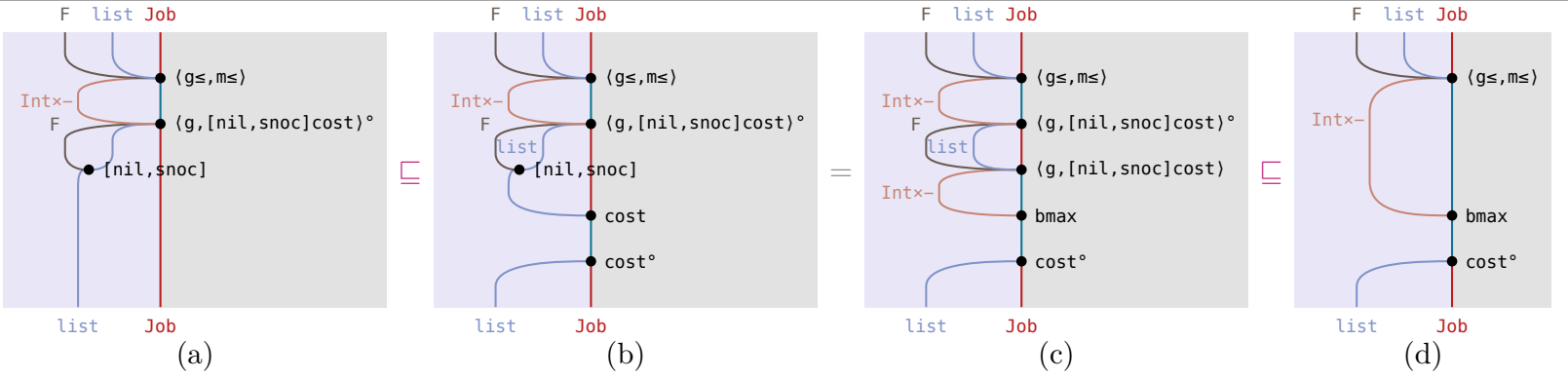
|     |                                                                                                                           |                                                                                |
|-----|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| (a) | $(Q \ n \ ([nil, snag] \ [nil, snag]^\circ)) \ F(bagify^\circ) \ [nil, snoc]$                                             | monotonicity of composition                                                    |
| (b) | $\sqsubseteq (Q \ F(bagify^\circ) \ [nil, snoc]) \ n \ ([nil, snag] \ [nil, snag]^\circ \ F(bagify^\circ) \ [nil, snoc])$ | $\beta^\circ F(bagify^\circ) \alpha = bagify^\circ$ , since $bagify = (\beta)$ |
| (c) | $= (Q \ F(bagify^\circ) \ [nil, snoc]) \ n \ ([nil, snag] \ bagify^\circ)$                                                | (10.3c)                                                                        |
| (d) | $\sqsubseteq (Q \ F(bagify^\circ) \ [nil, snoc]) \ n \ (F(bagify^\circ) \ m \leq cost^\circ)$                             | modular law                                                                    |



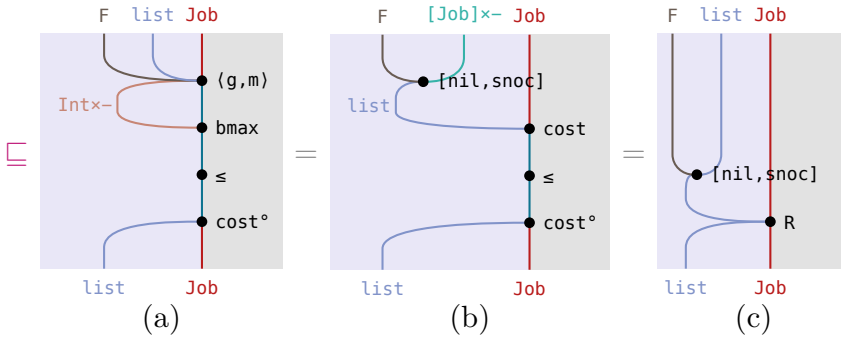
|     |                                                                                                                                 |                                                                                       |
|-----|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| (a) | $\sqsubseteq F(bagify^\circ) \ ((F(bagify) \ Q \ F(bagify^\circ)) \ n \ (m \leq cost^\circ \ [nil, snoc]^\circ)) \ [nil, snoc]$ | choice of $Q$ : $F(bagify) \ Q \ F(bagify^\circ) = g \leq g^\circ$                    |
| (b) | $= F(bagify^\circ) \ ((g \leq g^\circ) \ n \ (m \leq cost^\circ \ [nil, snoc]^\circ)) \ [nil, snoc]$                            | products: $\langle R, S \rangle \langle T, U \rangle^\circ = RT^\circ \ n \ SU^\circ$ |
| (c) | $= F(bagify^\circ) \ \langle g \leq, m \leq \rangle \langle g, [nil, snoc] \ cost^\circ \rangle^\circ \ [nil, snoc]$            | (10.3d)                                                                               |
| (d) | $\sqsubseteq F(bagify^\circ) \ [nil, snoc] \ R$                                                                                 |                                                                                       |

$$\langle g \leq, m \leq \rangle \langle g, [nil, snoc] \ cost^\circ \rangle^\circ \ [nil, snoc] \sqsubseteq [nil, snoc] \ R \quad (10.3d)$$

a job whose penalty is at most the last penalty, put after a schedule costing at most the cost before the last job, gives a schedule costing at most the whole



|     |                                                                                                                                                  |                                                  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| (a) | $\langle g \leq, m \leq \rangle \langle g, [nil, snoc] \ cost^\circ \rangle^\circ \ [nil, snoc]$                                                 | cost a map, so $1 \sqsubseteq cost \ cost^\circ$ |
| (b) | $\sqsubseteq \langle g \leq, m \leq \rangle \langle g, [nil, snoc] \ cost^\circ \rangle^\circ \ [nil, snoc] \ cost \ cost^\circ$                 | $\alpha \ cost = (g, \alpha \ cost) \ bmax$      |
| (c) | $= \langle g \leq, m \leq \rangle \langle g, [nil, snoc] \ cost^\circ \rangle^\circ \ \langle g, [nil, snoc] \ cost \rangle \ bmax \ cost^\circ$ | $\langle g, \alpha \ cost \rangle$ simple        |
| (d) | $\sqsubseteq \langle g \leq, m \leq \rangle \ bmax \ cost^\circ$                                                                                 | monotonicity of $bmax$                           |



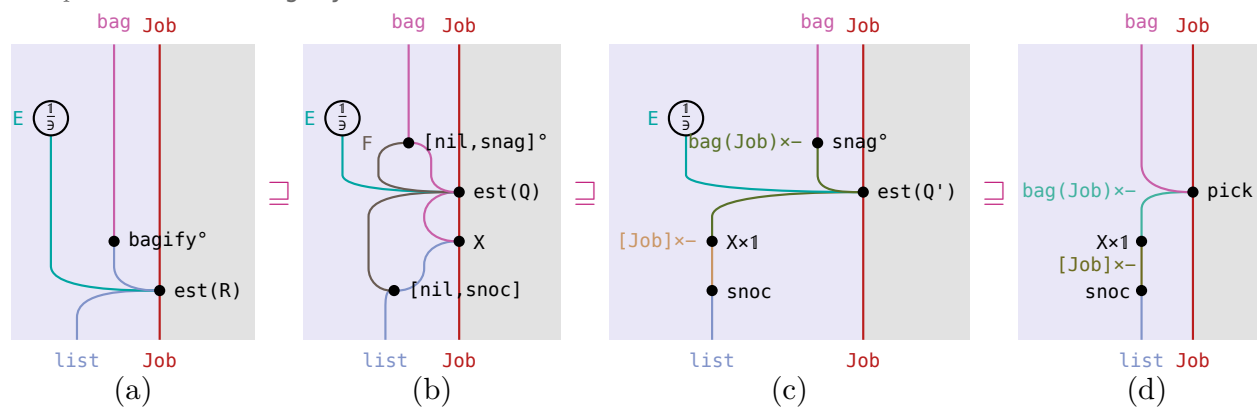
|     |                                                           |                   |
|-----|-----------------------------------------------------------|-------------------|
| (a) | $\sqsubseteq \langle g, m \rangle \ bmax \leq cost^\circ$ | (10.4) — (10.4)   |
| (b) | $= [nil, snoc] \ cost \leq cost^\circ$                    | definition of $R$ |
| (c) | $= [nil, snoc] \ R$                                       |                   |

$$\text{schedule} = (\text{null} \rightarrow \text{nil}, \text{pick} (\text{schedule} \times 1) \text{ snoc}), (\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (\mu X : \quad (10.3e)$$

$$\frac{[\text{nil}, \text{snag}]^\circ}{3} \text{ est}(Q) F(X) [\text{nil}, \text{snoc}] \sqsubseteq \frac{\text{bagify}^\circ}{3} \text{ est}(R)$$

an ordering of the given bag with least maximum penalty is got by taking a job of least penalty out of the bag, putting it last, and scheduling what is left the same way

the specification —  $\triangleq$  bagify



|     |                                                                                                       |                                                                                                                                                                    |
|-----|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{\text{bagify}^\circ}{3} \text{ est}(R)$                                                        | No greedy <b>reduce</b> exists — one would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole |
| (b) | $\sqsubseteq \frac{[\text{nil}, \text{snag}]^\circ}{3} \text{ est}(Q) F(X) [\text{nil}, \text{snoc}]$ | Proposition 10.1: <b>nil</b> and <b>snag</b> have disjoint ranges                                                                                                  |
| (c) | $\sqsubseteq \frac{\text{snag}^\circ}{3} \text{ est}(Q') (X \times 1) \text{ snoc}$                   | $\text{pick} \sqsubseteq \frac{\text{snag}^\circ}{3} \text{ est}(Q')$ , a partial function, quadratic in the number of jobs                                        |
| (d) | $\sqsubseteq \text{pick} (X \times 1) \text{ snoc}$                                                   |                                                                                                                                                                    |

## §10.4. The TeX problem

(10.4a)

$\text{interval} \triangleq \text{val round} : \text{Decimal} \rightarrow [0, 2^{16})$ ,  $\text{val} \triangleq ([\text{zero}, \text{shift}])$ ,  $\text{shift } (d, r) = (d+r)/10$ , round  $r$  rounds  $2^{16}r$  to the nearest integer:  $\text{round } r = n \iff 2n-1 < 2^{17}r < 2n+1$ .

$\text{interval } n = ((2n-1)/2^{17}, (2n+1)/2^{17})$ ,  $r \text{ inrange } (a, b) \iff a < r < b$ ,  $\text{round}^\circ = \text{interval inrange}$ ,  $R \triangleq \text{length} \leq \text{length}^\circ$ .

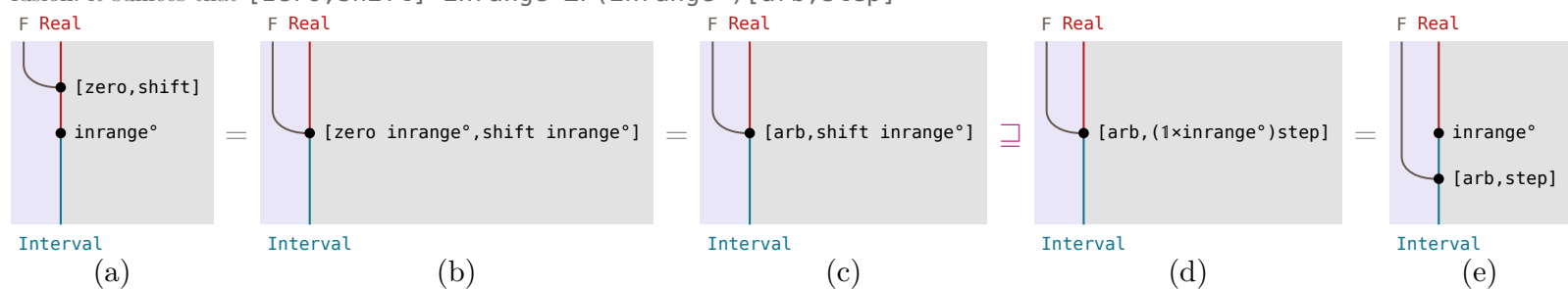
Interval the pairs  $(a, b)$  with  $\text{Legal}(a, b) \iff 0 < b < 1$  and  $a < b$  (10.9);  $\text{step\_legal}$  is what types  $\text{step } [\text{arb}, \text{step}] : 1 + (\text{Digit} \times \text{Interval}) \rightarrow \text{Interval}$ ,  $\text{step } (d, (a, b)) = ((d+a)/10, (d+b)/10)$ .

$\text{FX} = 1 + (\text{Digit} \times X)$ ,  $\alpha \triangleq [\text{nil}, \text{cons}]$ ,  $H \triangleq ([\text{arb}, \text{step}])^\circ$ ,  $! : \text{Digit} \times \text{Interval} \rightarrow 1$ ,  $Q \triangleq (l^\circ !^\circ r) \cup 1$   $l, r$  are coprod-laws's injections into  $\text{FX} = 1 + (\text{Digit} \times X)$ , so  $l : 1 \rightarrow \text{FX}$  and  $r : \text{Digit} \times X \rightarrow \text{FX}$ ,  $w \triangleq 2^{17}$ .

$$([[\text{arb}, \text{step}]] \sqsubseteq \text{val inrange}^\circ \quad (10.4b)$$

every interval got by folding a decimal's digits with  $[\text{arb}, \text{step}]$  has the decimal's value strictly inside it

fusion: it suffices that  $[\text{zero}, \text{shift}] \text{ inrange}^\circ \sqsubseteq F(\text{inrange}^\circ) [\text{arb}, \text{step}]$

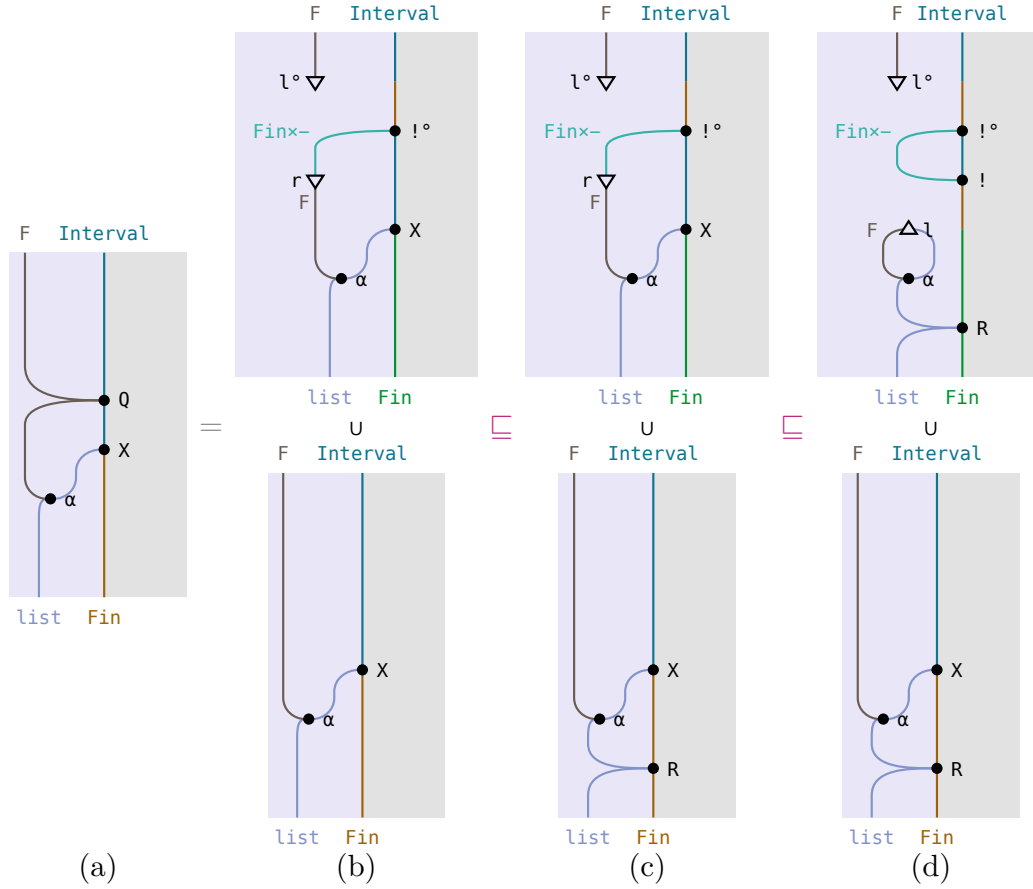


|     |                                                                          |                                                                                                                                                                                              |
|-----|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $[\text{zero}, \text{shift}] \text{ inrange}^\circ$                      | $[T, U]Z = [TZ, UZ]$ , a coproduct law not tabulated in the note                                                                                                                             |
| (b) | $= [\text{zero inrange}^\circ, \text{shift inrange}^\circ]$              | $\text{zero inrange}^\circ = \text{arb}$ : the first condition, which determines $\text{arb}$                                                                                                |
| (c) | $= [\text{arb}, \text{shift inrange}^\circ]$                             | arithmetic: $10a-d < r < 10b-d \implies a < (d+r)/10 < b$ for $(a, b) = \text{step}(d, (10a-d, 10b-d))$ ; only $\exists$ , since $(10a-d, 10b-d)$ satisfies (10.9) only when $d < 10b < d+1$ |
| (d) | $\sqsubseteq [\text{arb}, (1 \times \text{inrange}^\circ) \text{ step}]$ | $F(S) [T, U] = [T, (1 \times S) U]$ read right to left: definition of $F$                                                                                                                    |
| (e) | $= F(\text{inrange}^\circ) [\text{arb}, \text{step}]$                    |                                                                                                                                                                                              |

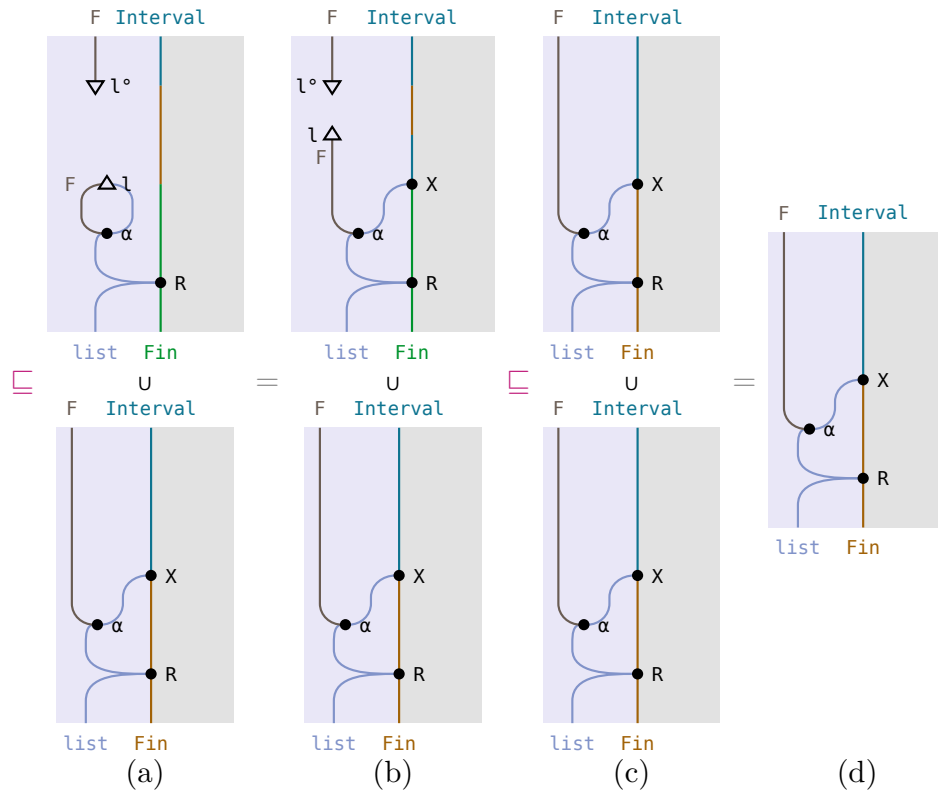
$$Q F(X) \alpha \sqsubseteq F(X) \alpha R$$

(10.4c)

choosing by  $Q$  among the one-step decompositions before building with  $F(X)$  and  $\alpha$  yields only decimals that are  $R$ -related to one built without choosing



|     |                                                                    |                                                                                                                         |
|-----|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| (a) | $Q F(X) \alpha$                                                    | $(S \cup T)U = SU \cup TU$ — definition of $Q$ ; composition distributes over $U$                                       |
| (b) | $= (l^\circ !^\circ r F(X) \alpha) \cup (F(X) \alpha)$             | $1 \sqsubseteq R$ — $R$ is reflexive                                                                                    |
| (c) | $\sqsubseteq (l^\circ !^\circ r F(X) \alpha) \cup (F(X) \alpha R)$ | $rF(X)\alpha \sqsubseteq !\alpha R$ — $\alpha = \text{nil}$ , and $\text{length}(\text{nil}) = 0$ is at most any length |
| (d) | $\sqsubseteq (l^\circ !^\circ ! l \alpha R) \cup (F(X) \alpha R)$  | $!^\circ ! \sqsubseteq 1$ on $1$ — universal property of $!$                                                            |

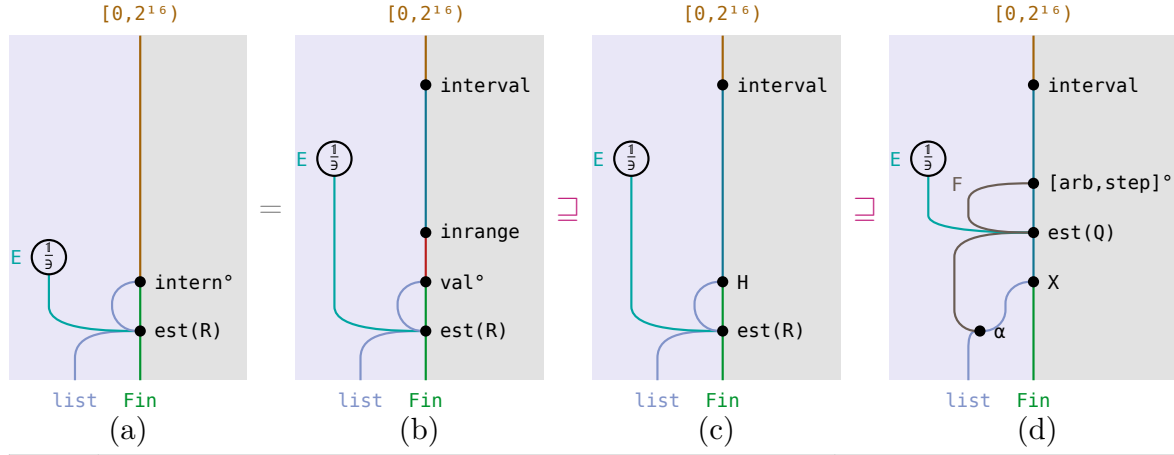


|     |                                                         |                                           |
|-----|---------------------------------------------------------|-------------------------------------------|
| (a) | $\sqsubseteq (l^\circ l \alpha R) \cup (F(X) \alpha R)$ | $lF(X) = l$ — definition of $F$           |
| (b) | $= (l^\circ l F(X) \alpha R) \cup (F(X) \alpha R)$      | $l^\circ l \sqsubseteq 1$ — $l$ is simple |
| (c) | $\sqsubseteq (F(X) \alpha R) \cup (F(X) \alpha R)$      | $S \cup S = S$ — $U$ is idempotent        |
| (d) | $= F(X) \alpha R$                                       |                                           |

$$\text{extern}(n)=f(2n-1,2n+1), \quad \text{interval } (\mu X : \frac{[\text{arb}, \text{step}]^\circ}{\exists} \text{est}(Q) F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{\exists} \text{est}(R) \quad (10.4d)$$

a shortest decimal whose internal representation is the given multiple of  $2^{-16}$  is got by emitting the one digit the interval of admissible reals allows, until that interval contains zero and the empty decimal will do

the specification —  $\triangleq \text{intern}$



|     |                                                                                                          |                                                                                                                                                                                          |
|-----|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{\text{intern}^\circ}{\exists} \text{est}(R)$                                                      | <code>round</code> is not a map, but <code>interval</code> is, so it comes out of the transpose                                                                                          |
| (b) | $= \text{interval } \frac{\text{inrange } \text{val}^\circ}{\exists} \text{est}(R)$                      | the type restriction (10.9): a shortest decimal <code>H</code> gives an interval is a shortest one among all decimals inside it, and is inside it by (10.4b)                             |
| (c) | $\sqsubseteq \text{interval } \frac{H}{\exists} \text{est}(R)$                                           | $\frac{[\text{arb}, \text{step}]^\circ}{\exists}$ returns at most two elements — stop, or take one more digit — and <code>! nil ⊆ cons R</code> makes it stop whenever stopping is legal |
| (d) | $\sqsubseteq \text{interval } \frac{[\text{arb}, \text{step}]^\circ}{\exists} \text{est}(Q) F(X) \alpha$ |                                                                                                                                                                                          |

- $f(a,b) = x \iff a < 0 \wedge x = [] \vee \neg a < 0 \wedge (\exists d. \exists y. d = [10b] \wedge f(10a-d, 10b-d) = y \wedge x = [d] \# y)$   
 $f$  gives  $(a,b)$  the empty decimal when  $a < 0$ , and otherwise the digit  $d = [10b]$  followed by what it gives  $(10a-d, 10b-d)$

(10.4e)

- $(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{\exists} \text{est}(Q) F(X) \alpha) = f$

the least solution of the greedy recursion of (10.4d) is  $f$

- $d = [10b] \wedge e = [10b] \implies d = e$

Exercise 10.12: two digits  $d$  and  $e$  with  $0 < 10b-d < 1$  and  $0 < 10b-e < 1$  are equal

- $(0 < q-p \wedge q < w) \wedge p < 0 \implies f(p,q) = []$

the program's first clause:  $f(p,q) = []$  when  $p < 0$

- $(0 < q-p \wedge q < w) \wedge \neg p < 0 \wedge d = 10q/w \implies f(p,q) = [d] \# f(10p-wd, 10q-wd)$

the program's second clause: with  $d = (10q) \text{ div } w$ ,  $f(p,q)$  is  $d$  followed by  $f(10p-w \cdot d, 10q-w \cdot d)$

- $(0 < q-p \wedge q < w) \wedge \text{Reach}(p,q) \implies f(p/w, q/w) = f(p,q)$

on the pairs the program reaches from `interval n`, the integer  $f$  at  $(p,q)$  is a value of the rational  $f$  at  $(p/w, q/w)$

- `interval f = extern`

$f$  after `interval` is the program `extern`, whose value at  $n$  is  $f(2n-1, 2n+1)$

## §10. Appendix

### §10.1. $P(S) \text{ est}(R) = (\exists S) n(\epsilon \setminus (SR^\circ))$

(10.1a)

$(p, q)$  tabulates  $W \triangleq (\exists S) n(\epsilon \setminus (SR^\circ))$ , and  $y \triangleq \frac{(p \exists S) n(qR)}{\exists}$ , a map.

(10.1b)

$$\begin{array}{c} \boxed{\begin{array}{c} W \\ P(S) \text{ est}(R) \end{array}} \xleftarrow{p^\circ q = W, 1 \sqsubseteq y y^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \\ \boxed{\begin{array}{c} y^\circ q \\ \text{est}(R) \end{array}} \end{array}$$

(10.1c)

$$\begin{array}{c} \boxed{\begin{array}{c} y^\circ q \\ \exists \end{array}} \xleftarrow{f^\circ \cdot \neg f^\circ, \cdot \exists \neg \exists} \boxed{\begin{array}{c} q \\ (p \exists S) n(qR) \end{array}} \xleftarrow{\Delta \neg n, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ q \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} 1 \\ R \end{array}} \end{array}$$

R reflexive

(10.1d)

$$\begin{array}{c} \boxed{\begin{array}{c} \epsilon y^\circ q \\ R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \exists, \circ, \text{meet}} \boxed{\begin{array}{c} R^\circ q^\circ q \\ R^\circ \end{array}} \quad \text{q simple} \\ \\ \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \xleftarrow{\Delta \neg n, \cdot T \neg / T, \circ, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \end{array} \\ \\ \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \xleftarrow{\cdot \exists \neg \exists, \text{meet}} \boxed{\begin{array}{c} p^\circ p \exists S \\ \exists S \end{array}} \quad \text{p simple} \\ \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} p \exists \\ (p \exists) n(qRS^\circ) \end{array}} \xleftarrow{1 \sqsubseteq q q^\circ} \boxed{\begin{array}{c} q^\circ p \exists \\ RS^\circ \end{array}} \xleftarrow{\circ, T \cdot \neg T \setminus} \boxed{\begin{array}{c} W \\ \epsilon \setminus (SR^\circ) \end{array}} \quad \text{W's right half} \end{array}$$

### §10.2. $P(\text{est}(R)) \text{ est}(R) = P(\text{dom}(\text{est}(R))) \text{ union est}(R)$

(10.2a)

$$\begin{array}{c} \boxed{P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{dom}(\text{est}(R)) \text{ est}(R) = \text{est}(R)} \boxed{P(\text{dom}(\text{est}(R)) \text{ est}(R)) \text{ est}(R)} \\ \\ \xrightarrow{P \text{ a relator}} \boxed{P(\text{dom}(\text{est}(R))) P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{est-711}} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \end{array}$$

(10.2b)

$$\begin{array}{c} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ P(\text{est}(R)) \text{ est}(R) \end{array}} \xleftarrow{(10.1b) \text{ at } S := \text{est}(R), \Delta \neg n, T \cdot \neg T \setminus} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \\ \boxed{\begin{array}{c} \epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\text{count of } T \cdot \neg T \setminus} \boxed{\begin{array}{c} \exists (\exists n(\epsilon \setminus (\epsilon \setminus R^\circ))) \\ \exists (\exists n(\epsilon \setminus R^\circ)) \end{array}} \\ \\ \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} (\exists \exists) n(\epsilon \setminus (\epsilon \setminus R^\circ)) \\ \exists \text{est}(R) \end{array}} \xleftarrow{T(UnV) \sqsubseteq TUnTV, \cdot \exists \neg \exists, \text{est-73}} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \\ \xleftarrow{P(\text{dom}(\text{est}(R))) \sqsubseteq 1} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \xleftarrow{\epsilon \text{ lax natural}} \boxed{\begin{array}{c} \text{dom}(\text{est}(R)) \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \\ \\ \xleftarrow{\text{dom}(\text{est}(R)) \sqsubseteq \text{est}(R) \text{ est}(R)^\circ} \boxed{\begin{array}{c} \text{est}(R) \text{ est}(R)^\circ \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\text{est}(R)^\circ \sqsubseteq \epsilon, \epsilon \epsilon \text{ union} \sqsubseteq \epsilon} \boxed{\begin{array}{c} \text{est}(R) \epsilon \text{est}(R) \\ \text{est}(R) R^\circ \end{array}} \quad \text{UP of est} \end{array}$$

### §10.3. Shortest paths on a cylinder, on lists

(10.3a)

| definition                                                                                                         | type                                               | note                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| $F(A, X) = A + A \times X$                                                                                         | $\mathcal{A} \rightarrow \mathcal{A}$              | The base functor: one square of the new column, alone or in front of the path so far.                                                  |
| $L = \text{list}^+$                                                                                                | $\mathcal{A} \rightarrow \mathcal{A}$              | A path is a non-empty list of squares, one per column crossed.                                                                         |
| $\alpha$ the initial algebra                                                                                       | $F(\text{dCL}(L, E)) \rightarrow \text{dCL}(L, E)$ | $\alpha(5) = [5]$ and $\alpha(1, [5]) = [1, 5]$ : a path is started by one square, or extended by one.                                 |
| $N$<br>the n-tuple relator                                                                                         | $\mathcal{A} \rightarrow \mathcal{A}$              | One component per row of a column, so the fold carries n answers at once.                                                              |
| $R \triangleq \text{sum} \leq \text{sum}^\circ$                                                                    | $[\text{Nat}] \rightarrow [\text{Nat}]$            | The cost of a path, which the cheapest minimises.                                                                                      |
| setify                                                                                                             | $A[n] \rightarrow PA$                              | $\text{setify}(1, 2, 3, 4) = \{1, 2, 3, 4\}$ — which row a component came from is forgotten.                                           |
| moves                                                                                                              | $A[n] \rightarrow P(A[n])$                         | $\text{moves}(x) = \{\text{up}(x), x, \text{down}(x)\}$ — rotated up, unrotated, rotated down.                                         |
| trans                                                                                                              | $P(A[n]) \rightarrow (P A)[n]$                     | $\text{trans}\{(a, b, c), (x, y, z)\} = (\{a, x\}, \{b, y\}, \{c, z\})$ — component k of the result is the set of the k-th components. |
| zip                                                                                                                | $F(a[n]) \rightarrow (F a)[n]$                     | $\text{zip}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = ((1, \{[5]\}), (2, \{[6]\}), (3, \{[7]\}), (4, \{[8]\}))$ .          |
| $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$                                                               | $F(PA) \rightarrow PFA$                            | $\text{cp}(1, \{[5], [6], [8]\}) = \{(1, [5]), (1, [6]), (1, [8])\}$ .                                                                 |
| $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union}))$<br>$\text{zip } N(\text{cp}(F(A, -)) E(\alpha))$ | $F(NA, N(P(TA))) \rightarrow N(P(TA))$             | $\text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}))$ is worked out in (10.3.1b).                                           |
| $\text{paths} \triangleq (\text{gen}) \text{ setify union}$                                                        | $T(NA) \rightarrow P(TA)$                          | $\text{paths}[(1, 2, 3, 4), (5, 6, 7, 8)]$ is the union of (10.3.1b)'s four sets: 12 paths, 3 from each entry row.                     |
| the specification<br>$\text{paths est}(R)$                                                                         | $[\text{Nat}[n]] \rightarrow [\text{Nat}]$         | A cheapest path from the entry side to the exit side.                                                                                  |

§10.3.1.  $N(E(L A))$ 

```

1  5
2  6      [(1,2,3,4), (5,6,7,8)] : L N Nat
3  7
4  8

```

(10.3.1a)

$$(\text{gen})[(5, 6, 7, 8)] = (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$$

(10.3.1b)

$$\begin{aligned} \text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = \\ ( \{ [1, 5], [1, 6], [1, 8] \}, \\ \{ [2, 5], [2, 6], [2, 7] \}, \\ \{ [3, 6], [3, 7], [3, 8] \}, \\ \{ [4, 5], [4, 7], [4, 8] \} ) \end{aligned}$$

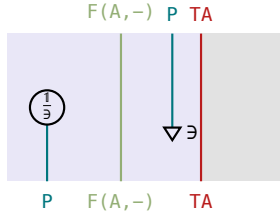
(10.3.1c)

|   |                                                        |
|---|--------------------------------------------------------|
| N | the 4-tuple, one component per row of the entry column |
| E | the set of paths that can start in that row            |
| L | a path is the non-empty list of the squares it crosses |

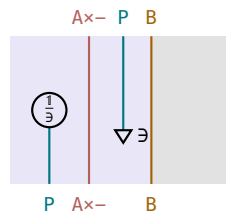
row 1's set has no  $[1,7]$  — from row 1 only rows 4, 1 and 2 are reachable — and it does have  $[1,8]$ , because the cylinder glues the bottom row to the top. That is **moves trans N(union)** of  $\triangleq$  **paths**: component  $k$  collects the paths of rows  $k-1, k, k+1$ .

### §10.3.2. The cross product $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$

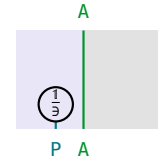
(10.3.2a)



cp on all of  $F(A, E(L A))$



the  $A \times -$  summand:  $\exists$  picks one path,  $1\% \exists$  collects the results



the  $A$  summand, no  $E$  to distribute:  $1\% \exists$  alone,  $a \mapsto \{a\}$

(10.3.2b)

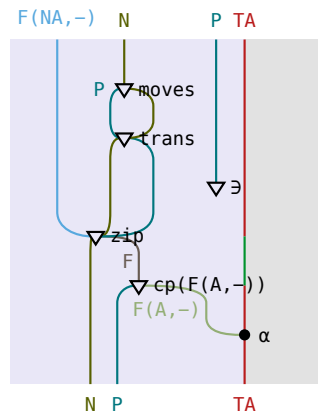
| type                                                              | note                                                                                                   |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| $F(A, -) = A + A \times -$                                        | one wire carries the whole functor, both summands with it; the $\exists$ under it is $F(1, \exists)$   |
| $\exists : E(L A) \rightarrow L A$                                | the $\exists$ inside $F(1, \exists)$ : a set of paths, one of them — $L A$ is one path, $A$ one square |
| $F(1, \exists) = 1 + 1 \times \exists$                            | $: A + A \times E(L A) \rightarrow A + A \times L A$ , the relator acting on each summand              |
| $\exists : E(A + A \times L A) \rightarrow A + A \times L A$      | the $\exists$ under the bar: a set of cells, one of them — a different $\exists$                       |
| $\text{cp} : A + A \times E(L A) \rightarrow E(A + A \times L A)$ | $\frac{R}{\exists}$ turns $R : X \rightarrow Y$ into $X \rightarrow E Y$                               |
| $\text{cp} = \frac{1}{\exists} E(F(1, \exists))$                  | what the picture draws: the unit makes the outer $E$ , leaving one $\exists$ — pow-laws                |

(10.3.2c)

$w = (1, \{[5], [6], [8]\}) : A \times E(L A)$  the right summand  
 $1 \times \exists : A \times E(L A) \rightarrow A \times L A$   $1$  keeps the square 1,  $\exists$  picks one path  
 $w \mapsto (1, [5]), (1, [6]), (1, [8]) : A \times L A$  one output per path in  $w$   
 $\text{cp}(w) = \{(1, [5]), (1, [6]), (1, [8])\} : E(A + A \times L A)$  the three of them, collected

### §10.3.3. $\text{gen} = F(1, \text{moves trans N(union)}) \text{ zip N(cp P}(\alpha))$

(10.3.3a)

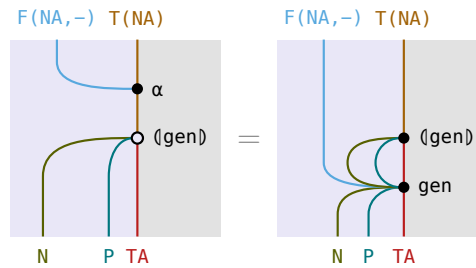


(10.3.3b)

$u = ((1,2,3,4),(\{[5]\},\{[6]\},\{[7]\},\{[8]\}))$  :  $F(N A, N(E(L A)))$   
 $F(1, \text{moves trans } N(\text{union}))$  : 1 keeps the column, the path SETS move  
 $\text{moves}(\{[5]\},\{[6]\},\{[7]\},\{[8]\})$   
 $= (\{[6]\},\{[7]\},\{[8]\},\{[5]\}),$   
 $(\{[5]\},\{[6]\},\{[7]\},\{[8]\}),$   
 $(\{[8]\},\{[5]\},\{[6]\},\{[7]\})$  :  $E(N(E(L A)))$  down, unmoved, up  
 $\text{trans}(\text{that})$   
 $= (\{[6]\},\{[5]\},\{[8]\},\{[7]\},\{[6]\},\{[5]\}),$   
 $(\{[8]\},\{[7]\},\{[6]\},\{[5]\},\{[8]\},\{[7]\})$  :  $N(E(E(L A)))$  row k gets rows k-1, k, k+1  
 $N(\text{union})(\text{that})$   
 $= (\{[5],[6],[8]\},\{[5],[6],[7]\},$   
 $\{[6],[7],[8]\},\{[5],[7],[8]\})$  :  $N(E(L A))$  every path into row k, none dropped  
 $\text{zip}(\text{that})$   
 $= ((1,\{[5],[6],[8]\}), (2,\{[5],[6],[7]\}),$   
 $(3,\{[6],[7],[8]\}), (4,\{[5],[7],[8]\}))$  :  $N(F(A, E(L A)))$  each row: its square, and the paths it may be put in front of  
 $N(\text{cp } P(\alpha))(\text{that})$   
 $= (\{[1,5],[1,6],[1,8]\},\{[2,5],[2,6],[2,7]\},$   
 $\{[3,6],[3,7],[3,8]\},\{[4,5],[4,7],[4,8]\})$  :  $N(E(L A))$  cp pairs the square with each path,  $\alpha$  prefixes it:  $\alpha(1,[5])=[1,5]$

#### §10.3.4. $\text{gen}$ is an F-algebra; $(\text{gen}): \alpha(\text{gen})=F(1, (\text{gen}))\text{gen}$

(10.3.4a)

 $\alpha$  puts the column back on the list, then the fold reads all of itthe fold reads the rest under  $F$ , then one  $\text{gen}$  puts the column in front

$$xs = [(1,2,3,4), (5,6,7,8)] = \alpha((1,2,3,4), [(5,6,7,8)]) : L(N A)$$

(10.3.4b)

$$(\text{gen})(xs) = \text{gen}((1,2,3,4), (\text{gen})[(5,6,7,8)])$$

both halves are in (10.3.1b): the tail folds to  $(\{[5]\},\{[6]\},\{[7]\},\{[8]\})$ , one path per row and each of them one square long, and  $\text{gen}$  on it is (10.3.3b)'s walk.

#### §10.3.5. $Q : F(N A, N(L A)) \rightarrow N(L A)$ , an $F(N A, -)$ -algebra: $Q=F(1, \text{moves trans } N(\text{est}(R))) \text{ zip } N(\alpha)$

(10.3.5a)

$Q : F(N A, N(L A)) \rightarrow N(L A)$  : the fold's algebra: a new column, one path per row  
 $u = ((1,2,3,4),([5],[6],[7],[8]))$  :  $F(N A, N(L A))$   
 $F(1, \text{moves trans } N(\text{est}(R)))$  : 1 keeps the column, the paths move  
 $\text{moves}([5],[6],[7],[8])$   
 $= ([6],[7],[8],[5]),$   
 $([5],[6],[7],[8]),$   
 $([8],[5],[6],[7]))$  :  $E(N(L A))$  down, unmoved, up  
 $\text{trans}(\text{that})$   
 $= ([6],[5],[8]),([7],[6],[5]),$   
 $([8],[7],[6]),([5],[8],[7]))$  :  $N(E(L A))$  row k gets rows k-1, k, k+1  
 $N(\text{est}(R))(\text{that})$   
 $= ([5],[5],[6],[5])$  :  $N(L A)$  the cheapest into each row – chosen BEFORE the new square  
 $\text{zip}(\text{that})$   
 $= ((1,[5]), (2,[5]), (3,[6]), (4,[5]))$  :  $N(F(A, L A))$  each row: its square, and the one predecessor that survived  
 $N(\alpha)(\text{that})$   
 $= ([1,5],[2,5],[3,6],[4,5])$  :  $N(L A)$   $\alpha(1,[5])=[1,5]$

| $Q=[N(\text{wrap}), (1 \times \text{moves trans } N(\text{est}(R))) \text{ zip}' N(\text{cons})]$ (10.3.5b)                                                                                                                                                                                                                                                                        |                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| at the last column $Q$ starts one path per row, and at each earlier one it puts each square in front of the cheapest of the three kept paths it can step to — the algebra (10.3.5c)'s last step folds                                                                                                                                                                              |                      |
| <b>circuit</b> — the fork is $F(NA, N(LA)) = NA + NA \times N(LA)$                                                                                                                                                                                                                                                                                                                 | <b>Hinze–Marsden</b> |
| $F(N A) (N (T A)) \text{---} \boxed{Q} \text{---} N(T A)$                                                                                                                                                                                                                                                                                                                          |                      |
| $FN(T A) \text{---} \boxed{F(1, \text{moves trans } N(\text{est}(R)))} \text{---} FN(T A) \text{---} \boxed{\text{zip}} \text{---} N(F A (T A)) \text{---} \boxed{N(\alpha)} \text{---} N(T A)$<br>the fusion condition (10.3.5d) read as a definition,                                                                                                                            |                      |
| $F(N A) (N (T A)) \text{---} \left( \begin{array}{c} \boxed{l} \text{---} N(p_0) \text{---} \boxed{N(\text{wrapA})} \\ \boxed{r} \text{---} q \text{---} \boxed{pm} \text{---} q \text{---} \boxed{\text{zip}'} \text{---} N(q_0) \text{---} \boxed{N(\text{consA})} \end{array} \right) \text{---} N(T A)$<br>$\text{zip} = 1 + \text{zip}', \alpha = [\text{wrap}, \text{cons}]$ |                      |

| <b>paths <math>\text{est}(R) \exists (Q)</math> setify <math>\text{est}(R)</math></b> (10.3.5c)                                                                                                                                                                                                                                                                                                                                              |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| shortest paths on a cylinder: the dynamic programming approach to exhaustive search allows a path of least cost to be found in $O(n \times m)$ time; $Q$ is (10.3.5b)'s algebra.                                                                                                                                                                                                                                                             |               |
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                      | Hinze–Marsden |
| $T(N\ A) \text{---} \boxed{\text{paths}} \text{---} P(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$                                                                                                                                                                                                                                                                                                                                |               |
| $= T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (P\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NP(T\ A) \text{---} \boxed{\text{setify}} \text{---} PP(T\ A) \text{---} \boxed{\text{union}} \text{---} P(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\triangleq \text{paths at paths}$                                                                                                                                       |               |
| $\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (P\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NP(T\ A) \text{---} \boxed{\text{setify}} \text{---} PP(T\ A) \text{---} \boxed{P(\text{est}(R))} \text{---} P(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $P(\text{est}(R))\ \text{est}(R) \sqsubseteq \text{union}\ \text{est}(R) \text{--- a minimum in each set, then a minimum of those; } R \text{ transitive}$ |               |
| $\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (P\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NP(T\ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T\ A) \text{---} \boxed{\text{setify}} \text{---} P(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\text{setify lax natural}$                                                                                                                                 |               |
| $\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (T\ A))} \text{---} \boxed{Q} \text{---} N(T\ A) \text{---} \boxed{\text{setify}} \text{---} P(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\text{fusion at (10.3.5d)}$                                                                                                                                                                                                      |               |

| <b>gen <math>N(\text{est}(R)) \exists F(1, N(\text{est}(R))) Q</math></b> (10.3.5d)                                                                                                |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| fusion: the condition for fusion in (10.3.5c)'s last step, used to derive a definition of $Q$ .                                                                                    |               |
| circuit                                                                                                                                                                            | Hinze–Marsden |
| $F(N\ A)\ (N\ (P\ (T\ A))) \text{---} \boxed{\text{gen}} \text{---} NP(T\ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T\ A)$                                               |               |
| $\sqsubseteq FNP(T\ A) \text{---} \boxed{F(1, N(\text{est}(R)))} \text{---} FN(T\ A) \text{---} \boxed{Q} \text{---} N(T\ A)$ $(7.13), \text{ then zip, trans, moves lax natural}$ |               |

(7.13) is  $F(1, \text{est}(R)) \alpha \sqsubseteq \text{cp } P(\alpha) \text{ est}(R)$ , mon-thm71 at the map  $\alpha$  with  $\frac{F(1, \exists) \alpha}{\exists} = \text{cp } P(\alpha)$ : extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.